

Logistic Regression: Pima Indians Diabetes Dataset

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2026-10-03

Regression

Regression is a broad statistical idea used to examine relationships between an outcome variable and one or more predictor variables.

Some common types of regression include:

- **Simple linear regression**
- **Multiple linear regression**
- **Logistic regression**

Logistic regression is used when the outcome variable has **two possible categories**, such as yes/no, success/failure, or diabetes/no diabetes.

Pima Indians Diabetes Dataset

In this example, we use the Pima Indians Diabetes dataset. The outcome variable indicates whether a person has diabetes.

We will use **glucose** and **age** as predictor variables.

Import the Data

First, we load the dataset from GitHub.

```
url <- "https://raw.githubusercontent.com/jbrownlee/Datasets/master/pima-indians-diabetes.data.csv"
PimaIndiansDiabetes <- read.csv(url, header = FALSE)
head(PimaIndiansDiabetes)
```

```
##   V1  V2 V3 V4  V5   V6    V7 V8 V9
## 1  6 148 72 35   0 33.6 0.627 50  1
## 2  1  85 66 29   0 26.6 0.351 31  0
## 3  8 183 64  0   0 23.3 0.672 32  1
## 4  1  89 66 23  94 28.1 0.167 21  0
## 5  0 137 40 35 168 43.1 2.288 33  1
## 6  5 116 74  0   0 25.6 0.201 30  0
```

Rename the Variables

The original dataset does not contain column names, so we assign meaningful names to the variables.

```
names(PimaIndiansDiabetes) <- c(
  "pregnant",
  "glucose",
  "pressure",
  "triceps",
  "insulin",
  "mass",
  "pedigree",
  "age",
  "diabetes"
)

head(PimaIndiansDiabetes)
```

```
##   pregnant glucose pressure triceps insulin mass pedigree age diabetes
## 1         6     148       72      35        0 33.6   0.627  50         1
## 2         1      85       66      29        0 26.6   0.351  31         0
## 3         8     183       64       0        0 23.3   0.672  32         1
## 4         1      89       66      23       94 28.1   0.167  21         0
## 5         0     137       40      35     168 43.1   2.288  33         1
## 6         5     116       74       0        0 25.6   0.201  30         0
```

Examine the Structure of the Data

We can use `str()` to examine the structure and data types of the variables.

```
str(PimaIndiansDiabetes)
```

```
## 'data.frame':   768 obs. of  9 variables:
## $ pregnant: int  6 1 8 1 0 5 3 10 2 8 ...
## $ glucose : int 148 85 183 89 137 116 78 115 197 125 ...
## $ pressure: int  72 66 64 66 40 74 50 0 70 96 ...
## $ triceps : int  35 29 0 23 35 0 32 0 45 0 ...
## $ insulin : int   0 0 0 94 168 0 88 0 543 0 ...
## $ mass     : num 33.6 26.6 23.3 28.1 43.1 25.6 31 35.3 30.5 0 ...
## $ pedigree: num 0.627 0.351 0.672 0.167 2.288 ...
## $ age      : int  50 31 32 21 33 30 26 29 53 54 ...
## $ diabetes: int   1 0 1 0 1 0 1 0 1 1 ...
```

Logistic Regression Model

We fit a logistic regression model using `glm()`.

The outcome variable is **diabetes**, and the predictor variables are **glucose** and **age**.

```

model <- glm(
  diabetes ~ glucose + age,
  data = PimaIndiansDiabetes,
  family = binomial()
)

summary(model)

##
## Call:
## glm(formula = diabetes ~ glucose + age, family = binomial(),
##      data = PimaIndiansDiabetes)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -5.912449   0.462620  -12.78  < 2e-16 ***
## glucose      0.035644   0.003290   10.83  < 2e-16 ***
## age          0.024778   0.007374    3.36 0.000778 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 993.48  on 767  degrees of freedom
## Residual deviance: 797.36  on 765  degrees of freedom
## AIC: 803.36
##
## Number of Fisher Scoring iterations: 4

```

Interpretation

Outcome

The outcome variable is **diabetes**, which indicates whether the person has diabetes.

Predictors

The predictors in the model are:

- **Glucose**
- **Age**

Results

The logistic regression results can be interpreted in terms of the **odds of diabetes**.

A positive coefficient for **glucose** indicates that higher glucose levels are associated with higher odds of diabetes, while holding age constant.

A positive coefficient for **age** indicates that increasing age is associated with higher odds of diabetes, after accounting for glucose.

Therefore, in this model, **glucose and age have positive associations with the odds of diabetes**.